

Marketing ROI & Budget Reallocation Strategy

Attribution vs Regression Analysis

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Business Analytics Capstone Project

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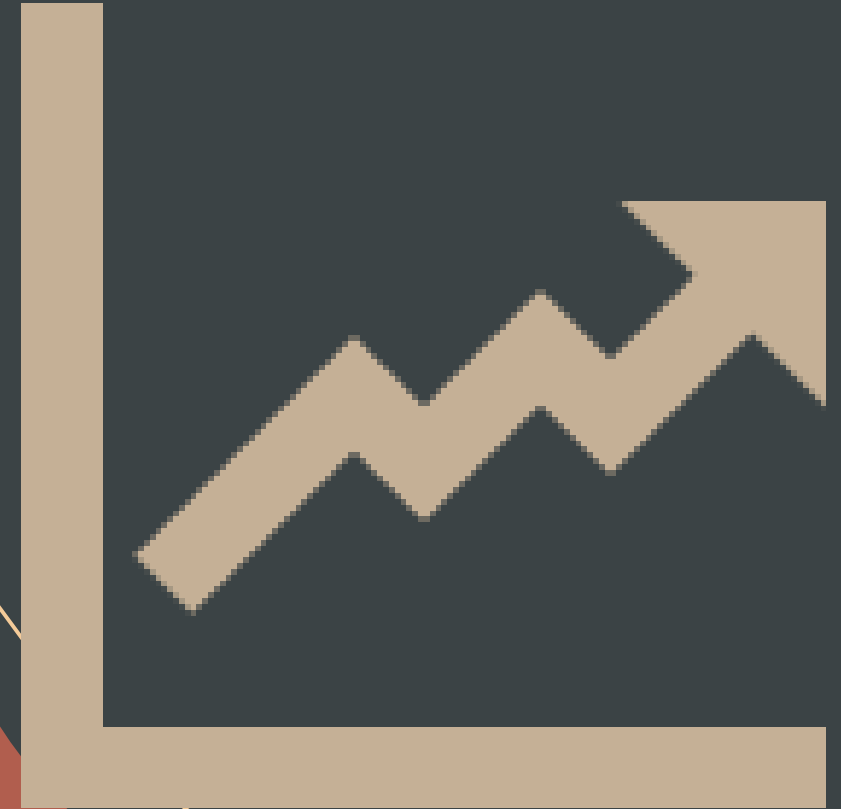


Business Objective

- The objective of this project is to evaluate marketing channel performance and identify opportunities to improve return on marketing investment (ROI). The analysis compares **last-touch attribution results with regression-estimated incremental impact** to determine whether current marketing budget allocation reflects the true contribution of each channel.

Key Questions

- Which channels generate the most revenue?
- Which channels are most efficient?
- Are we allocating marketing budget effectively?
- What changes could improve ROI in the next 30 days?



A stylized fingerprint graphic composed of several concentric, curved lines in a light beige color, set against a dark blue background. The lines are arranged to mimic the ridges of a fingerprint.

Data Sources & Analytical Approach

Data Sources

- fact_sessions – session-level user behavior
- fact_channel_daily – channel performance metrics
- fact_campaign_daily – campaign spend and revenue

Key Metrics

- $\text{ROAS} = \text{Revenue} / \text{Marketing Spend}$
- $\text{CAC} = \text{Marketing Spend} / \text{Customers}$
- $\text{Conversion Rate} = \text{Orders} / \text{Sessions}$
- Revenue per Session

Analytical Methods

- Last-Touch Attribution
- Regression Modeling (incremental impact estimation)

Marketing Performance Overview

- **Observations**
- Marketing generates strong revenue relative to spend.
- Search dominates revenue generation.
- Some channels show efficiency opportunities.

Metric	Value
Total Spend	\$74.47M
Revenue	\$102.11M
ROAS	1.37
CAC	\$29.46
Conversion Rate	2.02%

Key Insights from Channel Analysis



- Search generates the highest revenue and dominates conversion-stage activity.
- Email campaigns demonstrate strong efficiency with high conversion rates relative to spend.
- Paid Social receives significant budget allocation but exhibits lower ROAS and higher CAC.
- Organic traffic contributes meaningful revenue without direct marketing spend.
- New users show slightly higher conversion rates than returning users, indicating effective acquisition strategies.

Comparing Attribution and Regression Estimated Incremental Impact

Attribution Results

- Search receives **43% attribution credit**, making it appear to be the dominant driver of conversions.

Regression Results

- Regression modeling estimates Search's **incremental impact at ~30%**, suggesting last-touch attribution may over-credit bottom-funnel channels.

Key Takeaway

- Channels such as **Email, Paid Social, and Organic show higher incremental impact than attribution share**, indicating that they influence earlier stages of the customer journey.

Recommended Budget Reallocation

Strategy

- Increase investment in high-impact channels
- Reduce spending on lower-efficiency channels
- Maintain balanced marketing funnel coverage

Channel	Current Allocation	Proposed Allocation
Search	46.1%	55%
Paid Social	38.4%	20%
Email	5.1%	15%
Referral / Organic	10.4%	10%

Estimated Impact

- Expected 30-Day Impact
- Assuming a fixed monthly budget of **\$10.9M**, the proposed reallocation is expected to improve marketing efficiency.
- **Implication**
- Reallocating spend toward higher-impact channels could significantly improve ROI without increasing the overall marketing budget.

Scenario	Incremental Revenue
Base Case	+\$1.4M
Best Case	+\$2.2M
Worst Case	+\$0.5M



Risks & Limitations

Model limitations

- Regression identifies correlation rather than strict causality.

Attribution limitations

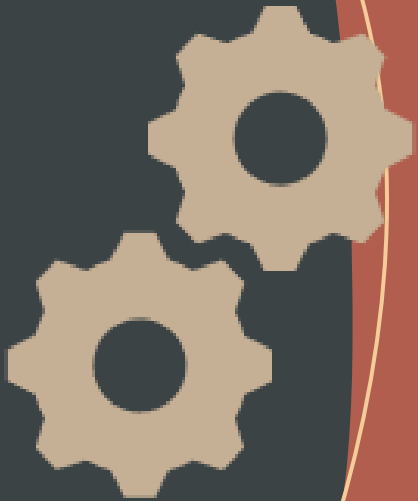
- Last-touch attribution ignores earlier marketing interactions.

Data limitations

- Limited visibility into full customer journey.

Operational risks

- Budget shifts may temporarily affect campaign performance.



Executive Recommendation: Optimize Budget Allocation for Higher ROI

Content

- Based on attribution and regression analysis, marketing performance can be improved by reallocating budget toward channels with stronger incremental impact and higher efficiency.

Key Recommendations

- **Increase investment in Search and Email**, which demonstrate strong conversion efficiency and incremental impact.
- **Reduce spend on Paid Social**, which currently consumes a large share of budget but shows lower ROAS and higher CAC.
- **Maintain baseline investment in Organic and Referral channels**, as they contribute meaningful revenue with minimal cost.
- Adopt a **data-driven budget allocation strategy** prioritizing channels with strong incremental impact to maximize marketing ROI.

Strategic Outcome

- Reallocating marketing spend toward higher-performing channels will improve overall marketing ROI while maintaining a balanced acquisition funnel.

Next Steps

- To further improve marketing decision-making:
 1. Implement **multi-touch attribution modeling**.
 2. Conduct **budget allocation experiments** to validate channel impact.
 3. Expand analysis to include **customer lifetime value and retention metrics**.
 4. Continuously monitor channel performance and optimize spend.